Project Design Phase-I
Problem – Solution Fit Template

Date	5 March 2026
Team ID	MCA_DA2
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns
Maximum Marks	2 Marks

Canvas Block	Our Answer
1. Customer Segment (CS)	Energy Policy Analysts at Ministry of Power State Electricity Board Administrators Data Analysts / BI Developers Academic researchers studying COVID-19 economic impact
2. Jobs-To-Be-Done / Problems	Compare state-wise consumption across 33 states for 2019 vs 2020 Identify which states were hit hardest by the COVID lockdown Track post-lockdown recovery (Jul–Dec 2020) by state and region Present findings visually to non-technical leadership
3. Triggers	Upcoming quarterly energy review meeting requiring state comparison Team lead asks for a shareable dashboard link instead of screenshots News reports about power demand collapse need data validation
4. Emotions Before / After	Before: Frustrated, overwhelmed, anxious about manual Excel errors After: Confident, efficient, recognised as the team's BI expert
5. Available Alternatives	Manual Excel pivot tables (slow, not shareable) Static POSOCO PDF reports (no interactivity) Power BI (paid license required) Google Data Studio (limited India map support)
6. Customer Constraints	Free tools only — Tableau Public, MySQL Community, Flask Non-technical stakeholders cannot install BI software Output must be shareable via URL — no file attachments
7. Behaviour	Analysts spend 2–3 hours/week manually building state comparison charts Share static screenshots via email and WhatsApp Rarely compare more than 5 states at once due to manual effort
8. Channels	Online: Tableau Public URL shared via email / Teams Online: Flask web app accessed via browser (no software needed) Offline: PDF exports and PowerPoint embeds for presentations
9. Problem Root Cause	POSOCO publishes weekly PDFs — no machine-readable interactive format No open dashboard exists for this specific 2019–2020 COVID dataset Raw CSV with 16,600 rows is unmanageable without a BI tool
10. Our Solution	MySQL + Tableau + Flask: 12+ visualisations, 3 dashboards, 5-scene story Pre/During/Post lockdown phase filter using SQL CASE WHEN Top N parameter, regional comparison, recovery tracking Published on Tableau Public — accessible via browser URL, zero installation